MSG No. 53.330 P_amna [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[9,10,13,14]`
4	$\left[0, 0, \frac{1}{2}\right]$	`[11,12,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[\frac{3}{4}, 0, 0\right]$	`[5,6,9,10]`
4	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[7,8,11,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,13,14]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[5,6,9,10]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,11,12]`

Table 5: Wyckoff site: `4e`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[0, y, \frac{1}{4}\right]$	`[1,6,11,16]`
2	$\left[0, -y, \frac{3}{4}\right]$	`[2,5,12,15]`
3	$\left[\frac{1}{2}, y, \frac{1}{4}\right]$	`[3,8,9,14]`
4	$\left[\frac{1}{2}, -y, \frac{3}{4}\right]$	`[4,7,10,13]`

Table 6: Wyckoff site: `4f`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, \frac{1}{4}\right]$	`[1,3,14,16]`
2	$\left[\frac{1}{4},\, -y,\, \frac{3}{4}\right]$	`[2,4,13,15]`
3	$\left[\frac{3}{4},\, -y,\, \frac{3}{4}\right]$	`[5,7,10,12]`
4	$\left[\frac{3}{4},\, y,\, \frac{1}{4}\right]$	`[6,8,9,11]`

Table 7: Wyckoff site: `8g`, site symmetry: `2..`

No.	position	mapping
1	$[x,\, 0,\, 0]$	`[1,2]`
2	$\left[\frac{1}{2}-x,\, 0,\, \frac{1}{2}\right]$	`[3,4]`
3	$[-x,\, 0,\, 0]$	`[5,6]`
4	$\left[x+\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[7,8]`
5	$\left[x+\frac{1}{2},\, 0,\, 0\right]$	`[9,10]`
6	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[11,12]`
7	$\left[\frac{1}{2}-x,\, 0,\, 0\right]$	`[13,14]`
8	$\left[x,\, 0,\, \frac{1}{2}\right]$	`[15,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	`[1,2]`
2	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,4]`
3	$\left[-x,\, \frac{1}{2},\, 0\right]$	`[5,6]`
4	$\left[x+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[7,8]`
5	$\left[x+\frac{1}{2},\, \frac{1}{2},\, 0\right]$	`[9,10]`
6	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[11,12]`
7	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, 0\right]$	`[13,14]`
8	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `m..`

No.	position	mapping
1	$[0,\, y,\, z]$	`[1,6]`
2	$[0,\, -y,\, -z]$	`[2,5]`
3	$\left[\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	`[3,8]`
4	$\left[\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	`[4,7]`
5	$\left[\frac{1}{2},\, y,\, z\right]$	`[9,14]`
6	$\left[\frac{1}{2},\, -y,\, -z\right]$	`[10,13]`
7	$\left[0,\, y,\, \frac{1}{2}-z\right]$	`[11,16]`
8	$\left[0,\, -y,\, z+\frac{1}{2}\right]$	`[12,15]`

Table 10: Wyckoff site: 8j, site symmetry: m’..

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, z\right]$	[1,14]
2	$\left[\frac{1}{4},\, -y,\, -z\right]$	[2,13]
3	$\left[\frac{1}{4},\, y,\, \frac{1}{2}-z\right]$	[3,16]
4	$\left[\frac{1}{4},\, -y,\, z+\frac{1}{2}\right]$	[4,15]
5	$\left[\frac{3}{4},\, -y,\, -z\right]$	[5,10]
6	$\left[\frac{3}{4},\, y,\, z\right]$	[6,9]
7	$\left[\frac{3}{4},\, -y,\, z+\frac{1}{2}\right]$	[7,12]
8	$\left[\frac{3}{4},\, y,\, \frac{1}{2}-z\right]$	[8,11]

Table 11: Wyckoff site: 8k, site symmetry: ..m’

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{4}\right]$	[1,16]
2	$\left[x,\, -y,\, \frac{3}{4}\right]$	[2,15]
3	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{4}\right]$	[3,14]
4	$\left[\frac{1}{2}-x,\, -y,\, \frac{3}{4}\right]$	[4,13]
5	$\left[-x,\, -y,\, \frac{3}{4}\right]$	[5,12]
6	$\left[-x,\, y,\, \frac{1}{4}\right]$	[6,11]
7	$\left[x+\frac{1}{2},\, -y,\, \frac{3}{4}\right]$	[7,10]
8	$\left[x+\frac{1}{2},\, y,\, \frac{1}{4}\right]$	[8,9]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[x,\, -y,\, -z]$	[2]
3	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$[-x,\, y,\, z]$	[6]
7	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[8]
9	$\left[x+\frac{1}{2},\, y,\, z\right]$	[9]
10	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	[10]
11	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\, -y,\, -z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, y,\, z\right]$	[14]
15	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[16]